



# Alpha Diallo

📍 Switzerland

🔗 alphadiallo.ch

✉ contact@alphadiallo.ch

🌐 alpha-diallo-09

## EXECUTIVE SUMMARY

Research scientist and software engineer leveraging years of experience managing IT teams and building robust software. With a proven track record in programming, data analytics, cloud computing, project management, and R&D (Research and Development), I have led complex IT projects, optimised data workflows, and fostered stakeholder partnerships.

## EDUCATION

**HEC Lausanne**  03/2020 – 02/2025  
**PhD, Information Systems** Lausanne, Switzerland

**University of Neuchâtel**  09/2017 – 12/2019  
**MSc, Information Systems** Neuchâtel, Switzerland

## WORK EXPERIENCE

**Romande Energie**  *Integration Engineer / Product Owner Middleware* 03/2025 – Present  
Morges, Switzerland

- Currently working on ensuring secure, efficient, and reliable communication between core components of the Information System
- Currently driving the adoption of best practices across cross-functional teams
- Defined and implemented an API governance strategy

**University of Lausanne**  09/2025 – 02/2026  
Lausanne, Switzerland

*Lecturer*

- Deliver weekly lectures to cohorts of up to 250 first-year Bachelor students, making algorithms, discrete mathematics and programming accessible across diverse academic backgrounds.
- Coordinate and mentor a team of 7 teaching assistants to ensure high-quality lab sessions and consistent student support.
- Design and adapt course materials that bridge computer science principles with forensic science, reinforcing the program's interdisciplinary relevance.

*Research Scientist* 03/2020 – 02/2025  
Lausanne, Switzerland

- Developed a platform for capturing and simulating human mobility in indoor environments (mobixim.unil.ch )
- Taught Programming, Algorithms and Computational Thinking to over 300 students each year
- Managed a team of up to six assistants per year
- Built a mobile application for collecting wireless and inertial measurements
- Supervised three master's theses on software engineering
- First-authored five papers published in selective computer science venues

*Student Assistant – Part-time* 11/2018 – 02/2020  
Lausanne, Switzerland

- Assisted students in building software by following programming best-practices in a Master's course on Software Architecture and a Bachelor's course on Algorithms and Computational Thinking

**Matchmore SA**  *Business Analyst – Part-time* 06/2018 – 10/2018  
Lausanne, Switzerland

- Developed three white-label applications to facilitate the usage of APIs and SDKs
- Drafted documentation and technical articles on Location-based Publish-Subscribe

- Acted as an intermediary between the technical and the business teams

**Infomaniak SA** , *Technical and Administrative Support Agent – Part-time*

- L1 and L2 technical and administrative support with a 100% satisfaction rate
- Live assistance for all the company's products (Web and Email, cloud servers/VPS, etc.)
- Server configuration and maintenance of web-based applications

03/2018 – 08/2018  
Geneva, Switzerland

**Private Investment Promotion Agency** , *Consultant, IT Manager – Full-time*

- Restructured and organized the IT department
- Implemented IT strategies and data governance policies, ensuring compliance with World Bank and IFC standards
- Led the development of a national portal of investments (invest.gov.gn )
- Created a platform offering investors reliable information on the business climate
- Devised a solution for collecting, storing and analysing data of up to 5000 private companies
- “Employee of the Month” in March 2015 for outstanding project leadership

12/2014 – 11/2016  
Conakry, Guinea

## LANGUAGES

French — C2

English — B2

German — A1

## SKILLS

|                                                                                                                                                                                   |                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Engineering and Analytics</b><br><i>MySQL, PostgreSQL, Oracle, MongoDB, DataOps, Statistical Programming, Data visualization, GIS and Spatial Analysis, ETL pipelines</i> | <b>Infrastructure and Services</b><br><i>Apache Kafka, RabbitMQ, on-premise MQTT Brokers, GCP, AWS, Linux SysAdmin, Cloud Data Architecture, DevOps, CI/CD, Containerization (Docker, Kubernetes)</i> |
| <b>Programming</b><br><i>Java (Springboot, JavaEE, JavaFX), Javascript (ReactJS, JQuery, VanillaJS), Python (Django, Flask, FastAPI), C/C++, Bash, C#</i>                         | <b>Project Management</b><br><i>Agile and Scrum Methodologies, Jira, GitHub Projects, Trello, Notion</i>                                                                                              |
| <b>Soft skills</b><br><i>Reliability, Work ethic, Flexibility, Professionalism</i>                                                                                                | <b>Machine Learning and Data Science</b><br><i>NumPy, Scikit-Learn, SciPy, TensorFlow, PyTorch, Keras, NLTK</i>                                                                                       |

## SELECTED PUBLICATIONS

**A Framework for Devising, Evaluating and Fine-tuning Indoor Tracking Algorithms**, *21st International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services (MobiQuitous 2024)*, Springer Nature  
Oslo, Norway

**Drift Correction of Inertial Indoor Tracking using Peer Collaboration and Strategic Beacons Placement**, *13th International Conference on Indoor Positioning and Indoor Navigation (IPIN 2024)*, IEEE  
Hong Kong, SAR China

**A Pragmatic Trade-off Between Deployment Cost and Location Accuracy for Indoor Tracking in Real-Life Environments**, *14th International Conference on Localization and GNSS (ICL-GNSS 2024)*, IEEE  
Antwerp, Belgium

**Decentralized Collaborative Inertial Tracking**, *20th International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services (MobiQuitous 2023)*, Springer Nature  
Melbourne, Australia

**Mobixim: A Framework for Devising Collaborative Algorithms**, *12th International Conference on Indoor Positioning and Indoor Navigation (IPIN 2023)*, CEUR Workshop Proceedings  
Nuremberg, Germany

## INTERESTS

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### Reading

Economics, Geopolitics, Anthropology

### Robotics and Internet of Things (IoT)

Embedded systems, CAD design, 3D printing

## VOLUNTEERING

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### Powercoders, *Coding Academy for Refugees*

Taught programming to refugees to facilitate their integration into Swiss society

2018 – 2020

Lausanne, Switzerland

### Hour of Code

Taught programming to children from underprivileged areas

2015 – 2016

Conakry, Guinea